Spacecraft Telemetry Health Assessment

Technical Assessment Report

> **System Concept:** Real-time decision support pipeline combining per-pass spacecraft telemetry with operator free-text notes. The system yields severity-graded health assessments with an auditable reasoning trace, requiring human operator confirmation for every action.

- **Pipeline Flow:** Telemetry + Note → Hard-limit Pre-filter → Numeric Scorer (Isolation Forest) + Text Scorer (Keywords) → Late Fusion → Operator Decision.
- **Dataset:** 109 synthetic labelled passes (60 nominal, 49 anomalous) spanning five subsystems (EPS, TCS, AOCS, Comms, OBC).
- **Reproducibility:** Completely deterministic training (seed 42) with reproducible pipeline scripts.

Approach and trade-offs

The pipeline utilizes a hybrid rule-based and machine learning architecture:

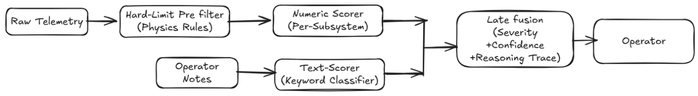

Figure 1: Pipeline Architecture and Information Flow

1.1 Hard limits as an independent pre-filter. Five fixed physics rules (battery voltage floor, component operational temperatures, tumble rate, RSSI-loss duration, and SEU error spikes) are evaluated *before* any learned model. Firing a rule flags the pass as CRITICAL at 1.0 confidence, mimicking standard out-of-limits (OOL) monitoring. This cannot be overridden by reassuring operator notes or downstream models.

1.2 Late fusion vs. early fusion. Numeric and text severities are computed fully independently and combined by an explicit fusion rule: take the higher severity, and escalate one band (e.g. WATCH to CAUTION) if both modalities independently agree at a non-trivial level. Fused confidence is the *minimum* of the two modalities, reflecting that "we are only as confident as our least-confident input." Early fusion (concatenating numeric + text features into a single classifier) was rejected because positive operator notes can mask genuine telemetry anomalies in a shared feature space.

1.3 Windowed vs. point-in-time scoring. Each subsystem is represented by summary statistics over the full 300-second pass window (mean, std, min, max, end-of-pass value, and rate-of-change). Rate-of-change features allow the system to detect slow drifts that stay within normal limits at any single point in time. Point-in-time models are structurally blind to trends and suffer from false-positive saturation when max-aggregating readings.

1.4 Classical anomaly detection. Per-subsystem Isolation Forests (primary) and One-Class SVMs (comparison) are trained on nominal-only data. This represents a natural fit for spacecraft monitoring, where nominal flight data is abundant but annotated anomaly cases are scarce. Deep learning models (e.g. LSTM-Autoencoders) add opacity and training cost without performance gains at this scale (~100 passes).

1.5 Keyword-based text classifier. Operator notes are classified using a hand-built keyword/phrase matching system that assigns weights to concern-indicating and reassurance-indicating phrases. This provides a direct, auditable reasoning trace for text scores, which is crucial for decision support.

Data and its limitations

There is no public annotated per-pass spacecraft health-telemetry corpus. Therefore, data was simulated for a representative LEO spacecraft class with a single-string EPS. (Note: We made a design assumption on the type of satellite, modeling parameter ranges based on a representative small satellite platform; these physical bounds are platformdependent and would vary for other classes). Each pass covers 300 readings at 1Hz across five subsystems. Physical channels are modeled contextually (e.g. battery voltage discharges during eclipse and charges in sunlight; RSSI follows orbit geometry).

Telemetry Parameter Schema

The generated dataset contains 17 parameters across five subsystems. The table below lists all parameters, their nominal behaviors, units, and flight heritage/physics source:

Subsys	Parameter	Unit & Type	Nominal Behavior & Bounds	Source & Tier
EPS	`battery_voltage`	V (Float)	6.2–8.4 V; slowly drops ~0.5V in eclipse, rises in sun. Noise: $\sigma = 0.02$	AAC Clyde / Aalto-1 (Tier 1)
	`rail_3v3`	V (Float)	Regulated 3.3V rail. Noise: $\sigma = 0.005$	NASA system specs (Tier 1)
	`rail_5v0`	V (Float)	Regulated 5.0V rail. Noise: $\sigma = 0.008$	NASA system specs (Tier 1)
	`solar_current`	A (Float)	~1.5A in sun, ~0A in eclipse. Range: 0.0–3.0A	Design assumption (Tier 3)
	`charge_rate`	V/s (Float)	Calculated battery voltage gradient (derivative)	Computed (Tier 3)
TCS	`sun_panel_temp`	°C (Float)	−20 to +75°C; transient response to sun/shade transitions	MinXSS flight data (Tier 1)
	`shade_panel_temp`	°C (Float)	−16 to +17°C; shaded radiator plate response	MinXSS radiator (Tier 1)
	`battery_temp`	°C (Float)	−5 to +40°C; comfort zone (lags sun panel temp)	Standard Li-ion (Tier 2)
	`internal_temp`	°C (Float)	~25°C; stable internal component temperatures	DeMi thermal limits (Tier 1)
AOCS	`wheel_speed`	RPM (Float)	3000–8000 RPM; nominal drift around 4500 RPM	CADRE / UWE-3 (Tier 1)
	`pointing_error`	deg (Float)	0.003 to 1.0°; nominal error stays around 0.5°	CADRE / BC XACT (Tier 1)
	`gyro_rate`	deg/s (Float)	MEMS angular rate; nominal centered at 0.0 ± 0.3	AAC Clyde PG400 (Tier 1)
	`altitude_error`	deg (Float)	Attitude tracking deviation; nominal centered at ~0.3°	Derived specs (Tier 1)
Comms	`rssi`	dBm (Float)	−100 to −60 dBm; parabolic profile centered on mid-pass	UHF link budget (Tier 3)
	`data_rate`	kbps (Float)	1.0 to 1000 kbps; scales dynamically with RSSI signal	Standard UHF specs (Tier 1)
OBC	`cpu_load`	% (Float)	0–100%; nominal load varies slowly around 30–50%	Generic computing (Tier 3)
	`memory_occupancy`	% (Float)	0–100%; nominal load ~50% with slight upward pass trend	Generic computing (Tier 3)
	`seu_count`	Count (Int)	0 nominal; random 1–3 correctable SEUs per pass	Flying Laptop (Tier 1/3)

Anomaly Injection Taxonomy

A programmatic injector produces four anomaly families:

1. **Point Anomalies:** Sudden spikes/drops (e.g. EPS battery voltage drop of 1.0–1.5V for 3–10 readings, OBC SEU spike of 3–8 errors/reading, Comms RSSI dropouts).
2. **Contextual Anomalies:** In-range values that contradict physical context (e.g. EPS battery voltage discharging during sunlight, TCS sun panel temperature rising in eclipse).
3. **Trend Anomalies:** Gradual drifts (e.g. EPS battery voltage decline of ~1.2V, TCS internal component temp rise of ~30°C, AOCS wheel speed drift).

4. **Hard-Limit Breaches:** Breaches of survival envelopes to validate rule-based safety monitoring (e.g. battery drop below 6.0V, internal temp above 100°C, angular velocity above 10 deg/s, RSSI signal loss for >30% of pass, seu count spike of 10+ errors/reading).

A total of 109 passes are generated: 60 nominal and 49 anomalous (including 10 borderline/uncertain passes injected at a low severity scale of 0.3).

Honest data limitations. (a) It is synthetic: real telemetry will have correlated noise, drift, and pass-to-pass variation that the generator does not reproduce, so precision will be lower on real data. (b) Several operationally important ranges are Tier 3 assumptions and would need mission-specific calibration before any real use.

Results and Validation

Calibration Correction

Percentile-rank normalization originally led to in-distribution nominal samples averaging a 0.5 score per subsystem, causing worst-of-five aggregation to flag all nominal passes as CRITICAL. This was resolved by switching to **envelope-distance normalization**: nominal envelope distances map to ~0, and the score increases only in the anomalous tail.

Fusion Topology: Late vs. Early Fusion

Early fusion was evaluated using a 95-feature vector fed to a Logistic Regression model. The models were evaluated against key test case `FUSION-TEST-001` (a numeric battery anomaly paired with a reassuring note). Late fusion correctly preserved the CRITICAL warning, whereas early fusion masked the telemetry anomaly, misclassifying the pass as NOMINAL.

Topology	Key-Case Severity	Key-Case Note	Aggregate Accuracy
Late Fusion (Production)	**CRITICAL** (conf 0.66)	Reassuring	61.5%
Early Fusion (Comparison)	NOMINAL	Reassuring	**98.2%**

Granularity: Windowed vs. Point-in-Time

Point-in-time Isolation Forests (max-aggregated over 300 readings) saturate near CRITICAL on nominal passes due to noise accumulation. In contrast, windowed scoring produces a calibrated spread across trend anomalies, capturing subtle drifts (e.g. `TREND-TCS-001`, `BORDER-0006`) as WATCH/CAUTION tier anomalies.

	Windowed	Point-in-time
Severity on Subtle Drift	NOMINAL (0.24)	CRITICAL (0.96)

Deep Learning Comparison on EPS Numeric Layer

An LSTM-Autoencoder trained on 100 nominal passes was compared against classical models. The LSTM-AE frequently flagged nominal passes (low precision), showing that classical models are more robust at this scale.

Model	Precision	Recall	F1
Isolation Forest (Production)	0.700	0.778	0.737
One-Class SVM (Comparison)	0.750	1.000	**0.857**
LSTM-Autoencoder (Control)	0.273	1.000	0.429

Validation Metrics

Hard-Limits: Fired with 100% detection and attribution (15/15 passes) with 0 false alarms. **Numeric Models:** Precision was high, but recall varied. EPS (0.78), AOCS (0.83), and Comms (1.00) anomalies were successfully flagged. However, TCS (0.36) and OBC (0.17) anomalies were missed because short bursts or contextual thermal shifts get diluted in window statistics. **Borderline Cases:** 10 borderline passes (`severity_scale=0.3`) yielded low-to-medium confidence (none exceeded 0.70), and 3/10 fired the LOW confidence flag (≤ 0.40). No nominal passes triggered high-confidence warnings.

Subsystem	IF Precision	IF Recall	IF F1	OC-SVM F1
EPS	0.70	0.78	0.74	0.86
TCS	0.80	0.36	0.50	0.57
AOCS	0.71	0.83	0.77	0.52
Comms	0.80	1.00	0.89	0.84
OBC	0.33	0.17	0.22	0.55

What is still weak

1. **Synthetic Data Limitations:** The clean noise model does not reflect real flight hardware; precision will drop on actual flight data.
2. **Uneven Subsystem Recall:** TCS and OBC contextual anomalies are diluted within window statistics, escaping detection.
3. **Stateless Text Classifier:** Keyword scoring is stateless and lacks temporal context (cannot catch repetitive operator warnings).
4. **Uncalibrated Hard Limits:** Physics rule thresholds use generic parameters and require component-level heritage tuning.
5. **No Cross-Pass Context:** Pass scoring is fully isolated, rendering the system blind to multi-day gradual drifts.
6. **No Within-Pass Localization:** Identifying anomalous passes is possible, but pinpointing the exact timestamp of an anomaly still requires raw telemetry inspection.

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